



HWA CHONG INSTITUTION (COLLEGE SECTION)
2021 9744 H2 BIOLOGY
PRELIMINARY EXAMINATION PAPER 4 MARK SCHEME

QUESTION 1

- (a)(i)** Measure the diameter of one of the cylinders in P and calculate the curved surface area, to the nearest whole number. (use π as 3.14) [2]

$$\begin{aligned}\text{Diameter of P} &= 7 / 8 \text{ mm} \\ \text{Curved surface area} &= (3.14 \times 8) \times 20 \\ &= 502.4 \text{ mm}^2 \\ &= 440 / 500 / 502 \text{ mm}^2\end{aligned}$$

- 1 correct measurement of diameter
- 2 correct final answer to nearest whole number

- (ii)** For the potato cylinders in P, calculate the area of one circular end to the nearest whole number. (use π as 3.14) [1]

$$\begin{aligned}\text{Area of one circular end} &= 3.14 \times 3.5^2 / 4^2 \\ &= 50.2 \text{ mm}^2 \\ &= 38 / 50 \text{ mm}^2\end{aligned}$$

- 1 correct final answer to nearest whole number

- (iii)** Complete Table 1.3 to calculate the total surface area when using different numbers of pieces of potato cylinders. [2]

Table 1.3

Table for 8mm

number of pieces cut from one cylinder	number of circular ends	area of one circular end from (a)(ii) / mm ²	surface area of all the ends / mm ²	curved surface area from (a)(i) / mm ²	total surface area / mm ²
1	2	50	100	500 / 502	600 / 602
2	4		200		700 / 702
3 / 4	6 / 8		300 / 400		800 / 802 / 902
4 / 8	8 / 16		400 / 800		900 / 902 / 1302

Table for 7mm

number of pieces cut from one cylinder	number of circular ends	area of one circular end from (a)(ii) / mm ²	surface area of all the ends / mm ²	curved surface area from (a)(i) / mm ²	total surface area / mm ²
1	2	38	76	440	516
2	4		152		592
3 / 4	6 / 8		228 / 304		668 / 744
4 / 8	8 / 16		304 / 608		744 / 1048

- 1 appropriate number of pieces cut from one cylinder
- 2 correct calculated total surface area for all

- (iv) Describe how you will standardise the position of the delivery tube in the test-tube of water, as shown in Fig. 1.1. [1]

ref to marking a mark on boiling / delivery tube / AW

- (v) Record your results for the total surface as shown in Table 1.3 and the number of bubbles at each 30 seconds in an appropriate table. [5]

Total surface area / mm ²	Number of bubbles for every 30s			
	30	60	90	120
602	6	5	5	5
702	9	8	8	7
902	12	11	10	10
1302	16	15	15	14

- 1 correct heading for total surface area
- 2 correct heading for number of bubbles
- 3 correct time intervals
- 4 correct recording of bubbles
- 5 correct trend: increasing number of bubbles as surface area increases

- (vi) Using the results in (a)(v), calculate for the largest surface area:

- the mean number of bubbles in 30 seconds
- the rate of activity, number min⁻¹.

Show all the steps in your working and use appropriate units. [3]

$$\text{Mean number of bubbles in 30 s} = \frac{21+20+19+18}{4} = 19.5$$

$$\text{Rate of activity} = \frac{20}{30} \times 60 = 40 \text{ bubbles min}^{-1}$$

- 1 correct addition of results for largest surface area
- 2 correct division by number of results to find mean
- 3 correct calculation of rate of activity

- (vii) Complete Table 1.4 to suggest:

- one other significant source of error in this procedure
- how to make improvements to reduce these errors. [3]

Table 1.4

Sources of error		Improvement	
1	Different sizes of bubbles released	1	ref to use of a gas syringe / displacement of water method
2	ref to different amounts of catalase in different potatoes	2	ref to use of same potato ;

- (b)(i) State 5 suitable concentrations of mannitol you will prepare from the stock solution. [1]

0.2, 0.4, 0.6, 0.8, 1.0 mol dm⁻³

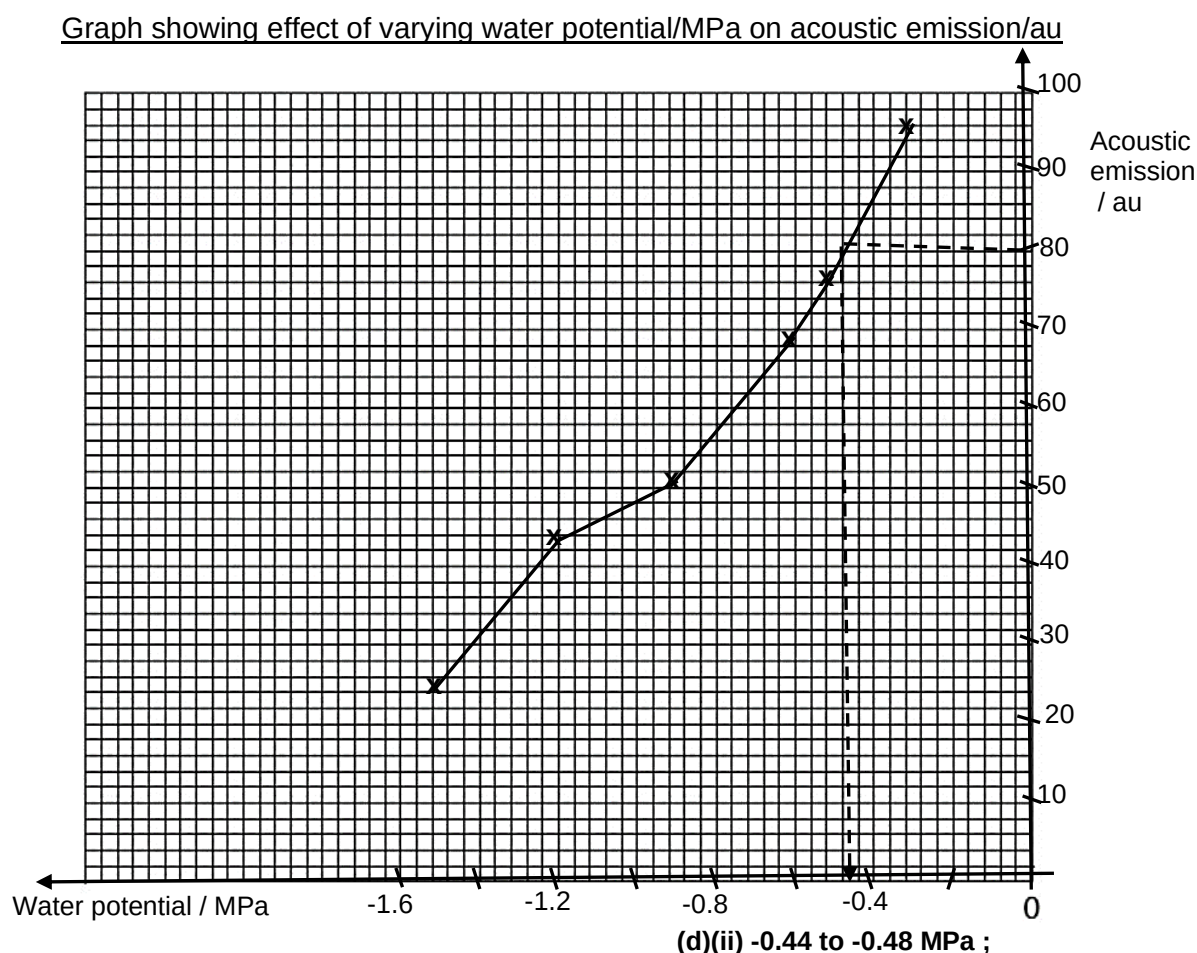
(ii) Using the materials and apparatus provided, design an experiment to estimate the water potential of potato tissue. *You do not have to carry out this experiment.* [5]

- 1 ref to cutting potato to appropriate size using cork borer, scalpel and mm ruler
- 2 ref to weighing potato sample to obtain initial mass using weighing balance
- 3 ref to correct set-up of experiment using mannitol solutions of varying concentrations and potato sample
- 4 ref to appropriate incubation time using stopwatch
- 5 ref to weighing potato sample to obtain final mass using weighing balance

(iii) Suggest how the results obtained from your experiment in (b)(ii) can be used to estimate the water potential of potato tissue. [2]

- 1 ref to determining concentration of mannitol which results in smallest percentage change in mass
- 2 ref to water potential of potato tissue is similar to the corresponding concentration of mannitol

(c)(i) Plot a graph of the data in Table 1.5 on the grid provided. [4]



- 1 correct axes labels and units
- 2 use of appropriate scales
- 3 correct plotting of points
- 4 point-to-point ruled line

(ii) Use your graph in (b)(i) to estimate the water potential of the potato tissue at an acoustic emission value of 80 arbitrary units. [1]

acceptable range of -0.44 to -0.48 MPa

[Total: 30]

QUESTION 2

- (a)(i) Use the space provided to draw a large plan diagram of the sector of the stem on **R1**, as shown by the shaded area in Fig. 2.1. [4]

A plan diagram shows the arrangement of different tissues, including their correct shapes and proportions. No cells should be drawn.

Use **one** ruled label line and the label **A** to identify an air space.

- 1 no cells + clear continuous lines + correct sector
 - 2 correct size of plan drawing + correct labelling of A
 - 3 correct arrangement of tissues
 - 4 correct shape + correct proportion
- (ii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, describe how you can determine the diameter of an air space. [2]
- 1 align EPG across the diameter of an air space
 - 2 count number of EPG units that span the diameter of an air space

- (iii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the mean diameter of an air space, in μm . [3]

Show the measurements that you made and your working.

- 1 show calibration of EPG division
 - 2 determine mean number of EPG divisions
 - 3 multiply mean number of EPG with length of one EPG
- (iv) The mean thickness of the cells in the sector has been determined to be 0.98 mm. Calculate the ratio of the mean diameter of the air space to the mean thickness of the cells. [2]
- 1 convert mm to μm
 - 2 display in simplest ratio

- (v) Within this sector of the stem there is a number of air spaces. Observe the cells that are found between two air spaces.

Select a group of four adjacent cells that are found between two air spaces. Each cell must touch at least two other cells.

Make a large labelled drawing of this group of four cells. [4]

- 1 clear continuous lines + cell wall shown as double lines
- 2 correct arrangement of cells
- 3 correct shape of cells
- 4 correct labels

- (vi) Identify observable differences between the stem on **R1** and the stem in Fig. 2.2. Record the observable differences in Table 2.1. [2]

Table 2.1

feature	R1	Fig. 2.2
1. central pith	absent	present
2. vascular bundle	arranged in two rings	arranged one ring
3. ring of air space	present	absent

(b)(i) Use the space provided below to record your results in a tally chart for a range of lengths from 1 mm to 12 mm, at 1-mm intervals. [2]

- 1 suitable column headings with appropriate units + tally marks
- 2 frequency + correct total count + correct range

(ii) Explain why all the long grains of rice are not exactly the same length. [2]

- 1 polygeny
- 2 environmental influence

(c)(i) Complete the final column of Table 2.3. [1]

Table 2.3

Observed frequency (O)	Expected frequency (E)	$O - E$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
1436	1476.5	- 40.5	1640.25	1.11
4470	4429.5	40.5	1640.25	0.370
$\chi^2 = \sum \frac{(O - E)^2}{E}$				1.48

(ii) Using the calculated χ^2 value from **(c)(i)** and Table 2.4, find the probability that the observed results deviated from the expected by chance. Explain how this probability is determined. [2]

- 1 $0.20 < p < 0.5$
- 2 $df = 1 + \chi^2$ value of 1.48

(iii) State what conclusion may be drawn from the probability found in **(c)(ii)**. [1]

$p > 0.05$ + no significant difference between observed and expected numbers

[Total: 25]